



UNIVERSITY OF PERADENIYA - FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER II) – 2020/2021
CSC1023 - OBJECT-ORIENTED PROGRAMMING (3 CREDITS)
/CS102 - PROGRAMMING TECHNIQUES (3 CREDITS)



Answer ALL Questions

Time allowed: Two Hours

1. [25 Marks]

- (a) Briefly describe abstraction using a real-world example. **03 Marks**
- (b) Briefly describe how the concept of abstraction benefits object-oriented programming. **03 Marks**
- (c) Explain the support of abstract methods for abstraction with a suitable Java programme. **06 Marks**
- (d) Describe two behavioural differences between interfaces and abstract classes in Java. **06 Marks**
- (e) “Cats and dogs are categorized as pets, while all pets are animals”.
Develop a Java programme structure (attributes and methods are not required) to demonstrate the relationship of classes. **07 Marks**

2. [25 Marks]

- (a) Describe a suitable Java programme where checking object equality is essential. **04 Marks**
- (b) Can two objects which are not equal have the same ‘Hash Code’? Explain your answer. **03 Marks**
- (c) Write the method signatures of `equals()` and `hashCode()` in Java. **04 Marks**
- (d) Develop a Java method that takes two `int` arrays and returns `true` if the arrays are disjoint (have no elements in common). **06 Marks**
- (e) Develop a Java method to check whether a given integer is a power of 2 or not using the bitwise AND operator (`&`). If the argument is the power of 2, it returns `true`, `false` otherwise.
Hint: If the number (n) is a power of 2 then $n \& (n - 1) == 0$ **08 Marks**

3. [25 Marks]

- (a) Use a suitable example to demonstrate how generics reduce runtime exceptions in Java. **05 Marks**
- (b) Briefly describe the term streams in Java. **03 Marks**
- (c) Demonstrate how the polymorphism feature is used in Java streams. **07 Marks**
- (d) Explain a critical circumstance to use byte streams over the character readers in Java. **04 Marks**

- (e) Write Java code snippets for the following requirements with proper exception management.
 - i) Append a text to an existing file named `data.txt`.
 - ii) Read a UTF-8 encoded text file named `encoded.txt`.

06 Marks

4. [25 Marks]

- (a) What is the main difference between software design patterns and software frameworks?
04 Marks
- (b) Write two advantages of the Static Factory Method design pattern. **04 Marks**
- (c) Implement the Static Factory Method design pattern using Java technologies. **10 Marks**
- (d) Explain a suitable Java `Callable` interface example where using `Runnable` interface is complicated. **07 Marks**
